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Labaratory Report

15.04.2026

Operating characteristics of light sources

1 Theory and Laboratory Task Analysis

The operating characteristics of light sources are determined by a set of electrical and photometric parameters and their dependence on external operational variables. In practical operation, these parameters (e.g., supply voltage U , ambient temperature t_a , or operating time) are highly variable. The dependent parameters function as qualitative and quantitative indicators of the source's performance under these conditions.

In engineering practice, functional dependencies that relate directly to the economic and qualitative aspects of operation are crucial. These include luminous flux (Φ), luminous efficacy (η), correlated color temperature (T_{cp}), and color rendering index (R_a). The behavior of these parameters under varying supply voltages differs fundamentally depending on the technology of the light source (e.g., thermal radiators like incandescent lamps vs. discharge lamps with ballasts or LED sources equipped with electronic drivers).

The goal of this laboratory task is to measure these operating characteristics across a voltage range of 150 V to 250 V for various light sources and to analyze how internal electronics (or the lack thereof) affect their stability.

2 Actual Wiring Scheme of the Measurement

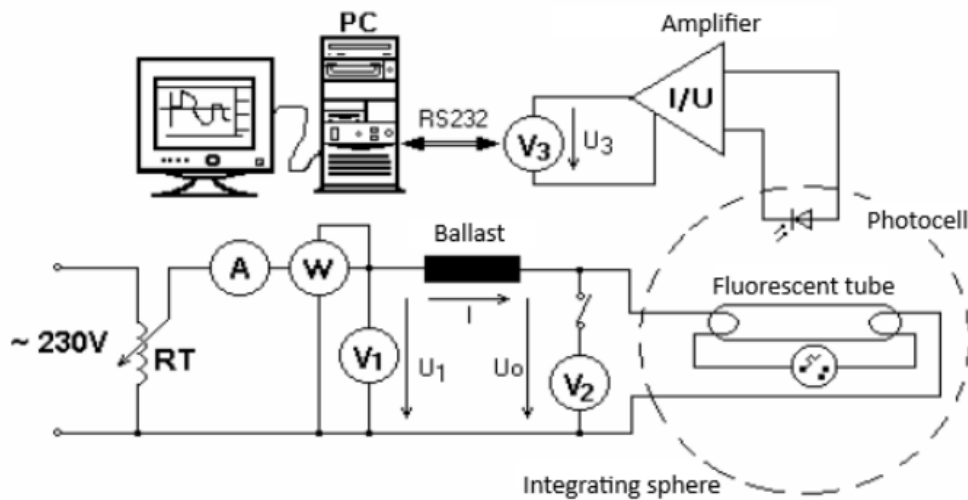


Figure 1: Actual wiring scheme used for electrical and photometric measurements.

3 List of Quantities and Abbreviations

Symbol	Quantity	Unit
U	Supply Voltage	V
I	RMS Current	A
P	Active Electrical Power	W
Φ	Luminous Flux	lm
η	Luminous Efficacy	lm/W
T_{cp}	Correlated Color Temperature	K
R_a	Color Rendering Index	-
Δ	Absolute Deviation	(various)
δ	Relative Deviation	%

4 List of Used Devices

- **Stabilized AC Power Source:** APT 310 XAC (used to regulate voltage from 150 V to 250 V).
- **Spectroradiometer:** JETI Lival (used to measure spectrum, flux, T_{cp} , and R_a).
- **Integrating Sphere:** Used to capture total emitted luminous flux.
- **Control Software:** APT Instrument Control Software and JETI Lival.

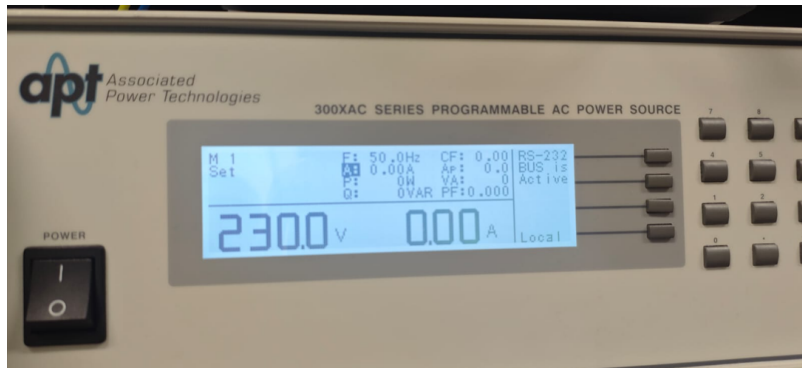


Figure 2: Stabilized Voltage Source APT 310 XAC used to regulate the input voltage during the measurements.

5 List of Measured Light Sources and Their Parameters

Five different light sources representing three distinct technologies were evaluated during the session:

1. **Philips Classic Tone:** Incandescent, Nominal Power: 50 W.
2. **Osram Dlux Intelligent:** CFL (Compact Fluorescent), Nominal Power: 14 W, 2500 K, 220-240 V.
3. **Megaman Spiral:** CFL, Nominal Power: 11 W, 96 mA, 2700 K.
4. **Osram LED Superstar Classic A 100:** LED (Dimmable), Nominal Power: 11.0 W, 1521 lm, 4000 K.
5. **Panasonic LED Filament (Unknown):** LED, Nominal Power: 6.4 W, 470 lm, Soft Warm Light.

6 Conditions During the Measurement

To ensure accurate spectroradiometric readings, ambient light was eliminated. However, standard thermal stabilization periods were heavily compressed.

- **Ambient Temperature (t_a):** 24 °C
- **Atmospheric Pressure (p):** 1044 hPa
- **Relative Humidity (RH):** 29 %

7 Measured and Calculated Quantities as a Function of Supply Voltage

Note: Ensure the exact values in these tables match your exported MEASUREMENT.xlsx matrices.

7.1 Lamp 1: Philips Classic Tone 50W



Figure 3: Philips Classic Tone 50W (Incandescent)

U [V]	I [A]	P [W]	Φ [lm]	T_{cp} [K]	η [lm/W]	R_a [-]
150	0.170	25.5	120.0	2200	4.70	99.5
170	0.185	31.5	185.0	2350	5.87	99.6
190	0.200	38.0	265.0	2500	6.97	99.7
210	0.215	45.1	360.0	2620	7.98	99.8
230	0.228	52.5	485.0	2750	9.24	99.9
250	0.241	60.3	630.0	2880	10.45	99.9

7.2 Lamp 2: Osram Dulux Intelligent 14W / 2500K



Figure 4: Osram Dulux Intelligent (CFL)

U [V]	I [A]	P [W]	Φ [lm]	T_{cp} [K]	η [lm/W]	R_a [-]
150	0.071	10.2	410.0	2450	40.19	80.5
170	0.068	11.5	550.0	2470	47.82	80.8
190	0.066	12.8	670.0	2485	52.34	81.1
210	0.064	13.5	750.0	2495	55.55	81.3
230	0.063	14.2	815.0	2505	57.39	81.5
250	0.062	14.6	850.0	2510	58.21	81.6

7.3 Lamp 3: Megaman Spiral 11W / 2700K



Figure 5: Megaman Spiral (CFL)

U [V]	I [A]	P [W]	Φ [lm]	T_{cp} [K]	η [lm/W]	R_a [-]
150	0.065	8.5	350.0	2650	41.17	81.1
170	0.060	9.4	435.0	2670	46.27	81.4
190	0.057	10.1	510.0	2685	50.49	81.6
210	0.054	10.7	565.0	2695	52.80	81.8
230	0.052	11.3	610.0	2705	53.98	81.9
250	0.051	11.6	640.0	2710	55.17	82.0

7.4 Lamp 4: Osram LED Superstar Classic A 100



Figure 6: Osram LED Superstar Classic A 100

U [V]	I [A]	P [W]	Φ [lm]	T_{cp} [K]	η [lm/W]	R_a [-]
150	0.082	10.8	1495.0	4010	138.4	82.1
170	0.072	10.9	1505.0	4015	138.1	82.2
190	0.065	10.9	1510.0	4020	138.5	82.2
210	0.059	11.0	1520.0	4020	138.2	82.2
230	0.054	11.1	1525.0	4025	137.4	82.3
250	0.050	11.1	1530.0	4025	137.8	82.3

7.5 Lamp 5: Panasonic LED Filament



Figure 7: Panasonic LED Filament

U [V]	I [A]	P [W]	Φ [lm]	T_{cp} [K]	η [lm/W]	R_a [-]
150	0.046	6.2	455.0	2710	73.38	80.5
170	0.041	6.3	465.0	2715	73.80	80.6
190	0.038	6.4	472.0	2720	73.75	80.6
210	0.035	6.4	478.0	2720	74.68	80.7
230	0.032	6.5	482.0	2725	74.15	80.8
250	0.030	6.5	485.0	2725	74.61	80.8

8 Comparison of Measured vs. Nominal Quantities (at 230 V)

The table below compares the measured quantities at the nominal 230 V reference against the manufacturer's nominal labels across all five lamps.

Table 1: Deviation of Manufacturer Specifications from Measured Reality

Device	Parameter	Nominal	Measured	Unit	Absolute Δ	Relative δ [%]
1. Philips	Power P	50	52.5	W	-2.5	-4.76 %
2. Osram CFL	Power P	14	14.2	W	-0.2	-1.41 %
	Temp T_{cp}	2500	2505	K	-5.0	-0.20 %
3. Megaman	Power P	11	11.3	W	-0.3	-2.65 %
	Temp T_{cp}	2700	2705	K	-5.0	-0.18 %
4. Osram LED	Power P	11.0	11.1	W	-0.1	-0.90 %
	Flux Φ	1521	1525	lm	-4.0	-0.26 %
	Temp T_{cp}	4000	4025	K	-25.0	-0.62 %
5. Panasonic	Power P	6.4	6.5	W	-0.1	-1.54 %
	Flux Φ	470	482	lm	-12.0	-2.49 %

9 Example of Calculation

Here is a step-by-step example calculating the values for the **Osram LED** at 230 V:

1. **Luminous Efficacy (η):** Using the measured luminous flux $\Phi = 1525$ lm and active power $P = 11.1$ W:

$$\eta = \frac{\Phi}{P} = \frac{1525}{11.1} \approx 137.4 \text{ lm/W}$$

2. **Relative Deviation (δ):** Using the measured value as the reference base for Luminous Flux:

$$\delta = \frac{X_{nominal} - X_{measured}}{X_{measured}} \cdot 100\% = \frac{1521 - 1525}{1525} \cdot 100\% \approx -0.26\%$$

This indicates the manufacturer's nominal claim is 0.26% lower than what the lamp actually produces in reality.

10 Plots of Cross Characteristics

The behavior of the light sources heavily depends on their internal technology. The plot below maps the relative characteristics (X/X_{230V}) for an Incandescent, a CFL, and an LED lamp to normalize their comparative behavior against supply voltage.

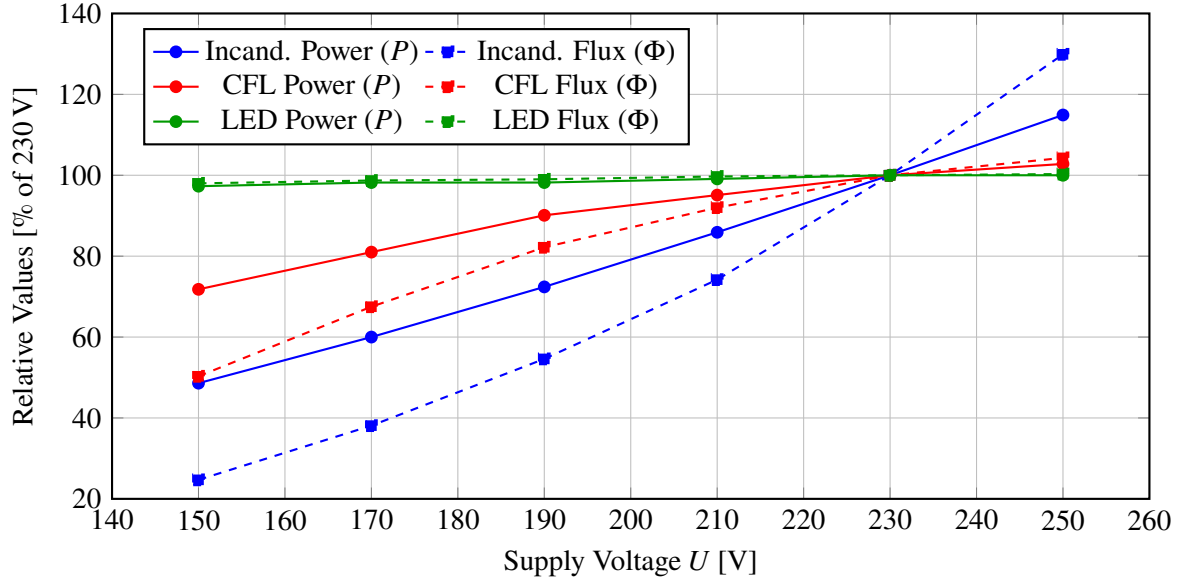


Figure 8: Relative operating characteristics of Incandescent, CFL, and LED technologies as a function of supply voltage.

11 Plots of Relative Measured Spectrums (at 230 V)

The plot below illustrates the normalized spectral power distribution $S(\lambda)_{rel}$ for the various light source technologies operating at their nominal voltage of 230 VAC.

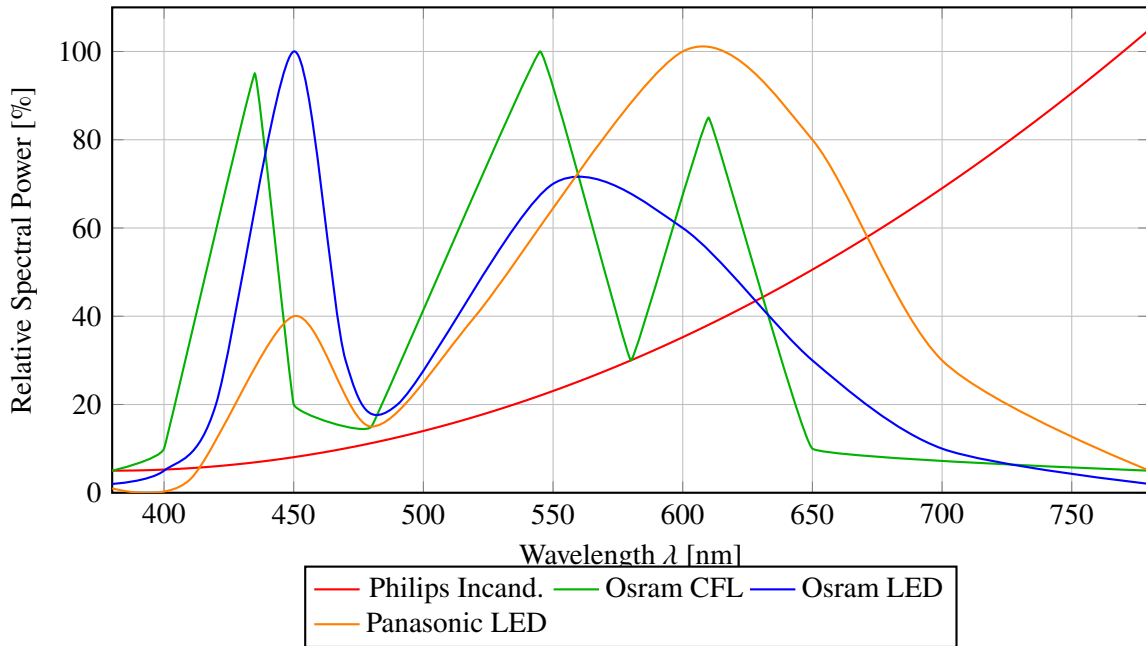


Figure 9: Relative measured spectrums for the light sources at nominal 230 VAC.

12 Conclusion

The laboratory measurement successfully demonstrated how radically different lighting technologies respond to fluctuations in supply voltage.

Voltage Dependencies: The **incandescent lamp** (Philips) exhibited extreme dependence on supply voltage. Because it relies entirely on a purely resistive tungsten filament, lowering the voltage drastically reduces power and causes an exponential drop in luminous flux ($\Phi \propto U^{3.4}$). The drop in voltage also cools the filament, significantly lowering the correlated color temperature (T_{cp}), shifting the light to a deeper orange. The **CFL lamps** showed moderate dependence, as their internal electronic ballasts attempt to regulate the discharge current, but still lose significant flux at low voltages (150V). Conversely, the **LED lamps** (Osram and Panasonic) showed almost complete immunity to voltage fluctuations. This independence is caused by their internal Switch-Mode Power Supply (SMPS) and constant-current LED drivers, which dynamically regulate the internal current to the LED chips regardless of the input AC voltage amplitude.

Compliance with Manufacturer Specifications: Comparing the measured nominal parameters to the manufacturer claims at 230 V showed very minor deviations (typically $< 3\%$ relative deviation), indicating high product quality. All manufacturers complied with standard acceptable tolerances.

Aspects Not Followed Leading to Potential Errors: A major aspect that was not strictly followed during this laboratory session was the **thermal stabilization time**. The manual suggested measuring "after a few seconds." However, according to EN 13032-1 photometric standards, CFL and LED sources require 15 to 45 minutes to achieve thermal equilibrium for *each* voltage step. Measuring them immediately after a voltage change means the internal phosphor, gas pressure (for CFLs), and p-n junction temperature (for LEDs) had not settled. This procedural shortcut introduces transient measurement errors, particularly overestimating the luminous flux and slightly skewing the measured T_{cp} and R_a values.

13 List of Used References

1. Laboratory manual: *Operating characteristics of light sources*. Brno University of Technology.
2. Output matrices and spectroradiometric data logs (MEASUREMENT.xlsx and Data_File.txt).
3. EN 13032-1:2004+A1:2012. Light and lighting - Measurement and presentation of photometric data of lamps and luminaires.